

# Momentum-transfer force rate: Orientation map

## Force from a steady stream of discrete momentum changes

**Subject:** Physics

**Topic:** Force and Laws of Motion

**Concept:** Force from a steady stream of discrete momentum changes

**Concept difficulty:** Hard concept

### Hard transition

A repeated stream converts a momentum change per item into a momentum change per unit time: if each fixed-mass item receives signed  $\Delta p_{\text{item}}$  and  $R$  such items are ejected per second, the average force on the ejecta is  $R \Delta p_{\text{item}}$ ; the launcher receives the opposite interaction force, so a stationary launcher requires an external holding force that balances that recoil.

### Governing relation

$$p = m v$$

The signed one-dimensional linear momentum of one fixed-mass item equals its mass times its signed velocity in the declared inertial frame.

- The item mass is positive and fixed during the compared states.
- Velocity and momentum use the same declared one-dimensional positive direction.
- Initial and final velocities are measured in the same inertial frame.

$$F_{\text{ext}} = dp/dt$$

The net external force on a chosen system equals its directed rate of change of linear momentum; this bounded Grade-9 extension evaluates that rate through a steady discrete transfer rather than calculus.

- System boundary, inertial frame and positive direction are declared.
- The momentum and force refer to the same chosen system and frame.
- For the learner route here, the rate is evaluated from repeated finite momentum transfers; no derivative calculation is required.

$$F_{\text{avg, on ejecta}} = R \Delta p_{\text{item}}$$

If identical fixed-mass items undergo the same signed momentum change and are ejected at steady rate  $R$  items per second, the ejecta stream's average momentum-change rate and average force equal  $R$  times the momentum change per item.

- The represented items are identical for the calculation or share the same  $\Delta p_{\text{item}}$ .

- $R$  is the number ejected per unit time over the averaging interval.
- $\Delta p_{\text{item}}$  uses the same inertial frame and positive direction for every item.
- The reported force is an average over the repeated discrete ejections.

### Compact anchor

Take the ejection direction as positive. Each identical item gains  $+2 \text{ kg m/s}$  of momentum, 5 items are ejected each second, and the launcher is held fixed. What are the average force on the ejecta, the recoil force on the launcher, and the external holding force?

$F_{\text{avg,on\_ejecta}} = R \Delta p_{\text{item}} = 5(+2) = +10 \text{ N}$ . The launcher receives the interaction-partner recoil force  $-10 \text{ N}$ , so a fixed launcher requires a  $+10 \text{ N}$  external holding force to balance that recoil.

### Before you go on

A launcher is held fixed. Identical items of mass  $m$  enter the launch process with signed velocity  $v_{\text{in}}$  and leave with signed velocity  $v_{\text{out}}$  in the same inertial frame. The steady ejection rate is  $R$  items per second, and positive is the exit direction. Give the average force on the ejecta, the interaction force on the launcher, the external holding force required to keep the launcher fixed, and the unit check.